# CS 180 Research Proposal

San Jose State University - Department of Computer Science
Fall 2026

## Title

Can you trust the explanation? Auditing the stability and agreement of feature-importance methods for cryptocurrency prediction models.

## Description

Machine-learning models are widely used to predict cryptocurrency price movements, and a growing number of papers pair these models with post-hoc explanation methods such as SHAP and LIME to report which features drive the predictions. Those reported feature-importance stories are typically presented as if they are solid facts about the market. This project asks whether they actually are. Explanation methods are known in the broader machine-learning literature to be unstable  sensitive to random seeds, sampling, feature correlation, and small input changes  yet this instability has not been carefully characterized in the specific, highly volatile setting of cryptocurrency prediction, where feature correlations are strong and noise is high.

The project builds standard crypto-prediction models on public data, computes explanations using several methods, and then systematically measures (a) how much the different methods agree on the top drivers, (b) how much those drivers shift when the model is retrained with a different seed or when inputs are slightly perturbed, and (c) the central question whether this instability is meaningfully worse during high-volatility market regimes than during calm ones. The expected and useful finding is a quantified answer to how far the feature-importance claims in this literature can be trusted, and under what conditions they break down.

The grade will be determined based on code submitted and a report produced describing the results.

### Scope and feasibility

- All data is public (exchange APIs / historical CSVs) and small trains on a laptop or modest GPU credits.
- Core stack (pandas, scikit-learn, XGBoost/LightGBM, shap, lime) is standard and quick to stand up.
- The grade-bearing novelty is the evaluation design and the honesty of the analysis, the part that requires the student's own judgment and cannot be delegated to tooling.

## Timeline

| Milestone | Action | Deliverable / Output | Reading |
|---|---|---|---|
| Week 1 | Read background on post-hoc explanation methods and their known instability; write the problem statement | 1-page summary + lit-survey outline | [1][2][3][4] |
| Week 2 | Assemble public crypto data (e.g. BTC, ETH, and 2-3 mid-cap coins); engineer standard features (lagged returns, RSI, MACD, volume, volatility) | Data + feature pipeline on GitHub | [9] |

| Milestone | Action | Deliverable / Output | Reading |
|---|---|---|---|
| Week 3 | Build a clean, strictly time-ordered train/test setup (walk-forward, no shuffling) to avoid the leakage flaw common in this literature | Documented validation harness on GitHub | [9] |
| Week 4 | Train the predictive models (XGBoost, LightGBM, and one small neural net) for next-day direction; record accuracy per regime | Trained models + performance table | [5] |
| Week 5 | Compute explanations (TreeSHAP, KernelSHAP, LIME, permutation importance) for each model. Submit design in UML | System design (UML) + explanation pipeline | [1][6][7] |
| Week 6 | Cross-method agreement: for each prediction, compare the top-k drivers across methods (Jaccard, rank correlation) | Agreement results + plots | [2][8] |
| Week 7 | Stability under retraining: retrain with different seeds and bootstrap samples; measure how much the top drivers move | Stability-under-retraining results | [2][4] |
| Week 8 | Stability under perturbation: nudge inputs within plausible ranges; measure how often the stated top driver changes | Perturbation-stability results | [3][4] |
| Week 9 | Regime split the key question: is instability worse in high-volatility periods than calm ones? Compare explicitly | Volatility-regime comparison | [2][9] |
| Week 10 | Consolidate across coins, models, and methods; build the main results tables and figures | Draft results section + figures | — |
| Week 11 | Repeated-run noise floor: quantify random variation so reported instability is real signal, not measurement noise | Robustness appendix on GitHub | — |
| Week 12 | Test cases for evaluation; verify the whole pipeline reproduces from a clean checkout | Summary of test cases and expected results | — |
| Week 13 | Make code and data available (features, seeds, environment file, method configs) | Repo + gDrive / Colab notebook | — |
| Week 14 | Code review | Recorded presentation of code and results | — |
| Week 15 | Write-up in a report | Report of results and conclusions on GitHub; workshop-paper draft | — |

## References


**[1]** Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems (NeurIPS) 30, 4765–4774. https://proceedings.neurips.cc/paper_files/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf

**[2]** Alvarez-Melis, D., & Jaakkola, T. S. (2018). On the robustness of interpretability methods. ICML 2018 Workshop on Human Interpretability in ML (WHI). arXiv:1806.08049. https://arxiv.org/abs/1806.08049

**[3]** Slack, D., Hilgard, S., Jia, E., Singh, S., & Lakkaraju, H. (2020). Fooling LIME and SHAP: Adversarial attacks on post-hoc explanation methods. AIES 2020, 180–186. DOI:10.1145/3375627.3375830. https://arxiv.org/abs/1911.02508

**[4]** Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. KDD 2016, 1135–1144. https://arxiv.org/abs/1602.04938

**[5]** Zhou, Z., Hooker, G., & Wang, F. (2021). S-LIME: Stabilized-LIME for model explanation. KDD 2021. https://arxiv.org/abs/2106.07875

**[6]** Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. KDD 2016, 785–794. https://arxiv.org/abs/1603.02754

**[7]** Pinky, J. N., & Akula, R. (2024). Enhancing cryptocurrency market forecasting: Advanced machine learning techniques and industrial engineering contributions. arXiv:2410.14475. https://arxiv.org/abs/2410.14475

**[8]** Ghorbani, A., Abid, A., & Zou, J. (2019). Interpretation of neural networks is fragile. AAAI 2019, 3681–3688. arXiv:1710.10547. https://arxiv.org/abs/1710.10547

**[9]** Molnar, C. (2022). Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd ed.). Free online book. https://christophm.github.io/interpretable-ml-book/